

IMO(1st) – 1959

First Day

1. Prove that the fraction $\frac{21n+4}{14n+3}$ is reducible for every natural number n .
2. For what real values of x is $\sqrt{(x+\sqrt{2x-1})} + \sqrt{(x-\sqrt{2x-1})} = A$, given (a) $A = \sqrt{2}$, (b) $A = 1$, (c) $A = 2$, where only non-negative real numbers are admitted for square roots?
3. Let a, b, c be real numbers. Consider the quadratic equation in $\cos x$: $a \cos^2 x + b \cos x + c = 0$. Using the numbers a, b, c form a quadratic equation in $\cos 2x$, whose roots are the same as those of the original equation. Compare the equations in $\cos x$ and $\cos 2x$ for $a = 4, b = 2, c = -1$.

Second Day

4. Construct a right triangle with given hypotenuse c such that the median drawn to the hypotenuse is the geometric mean of the two legs of the triangle.
5. An arbitrary point M is selected in the interior of the segment AB . The square $AMCD$ and $MBEF$ are constructed on the same side of AB , with the segments AM and MB as their respective bases. The circles circumscribed about these squares, with centers P and Q , intersect at M and also at another point N . Let N' denote the point of intersection of the straight line AF and BC .
 - (a) Prove that the points N and N' coincide.
 - (b) Prove that the straight lines MN pass through a fixed point S independent of the choice of M .
 - (c) Find the locus of the midpoints of the segments PQ as M varies between A and B .
6. Two planes, P and Q , intersect along the line p . The point A is given in the plane P , and the point C in the plane Q ; neither of these points lies on the straight line p . Construct an isosceles trapezoid $ABCD$ (with AB parallel to CD) in which a circle can be inscribed, and with vertices B and D lying in the planes P and Q .

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